MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON, TYPE 6344A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Magnetron, pulse, 5450 to 5825 MHz tunable frequency, 175 kw minimum peak power output, integral magnet, air cooled. Unilaterally interchangeable with 6344 magnetron in all systems.

ABSOLUTE -RATING: See Note 13

Independent:

Parameter:	if (surge)	tk	VSWR	Tuner Torque	Anode T	Bushing T	epy	Input Capacitance
Units:	a	sec	---	in-oz	°C.	°C	kv	pF
© Maximum:	---	---	1.5:1	160	150	175	24.0	25
Minimum:	---	180	---	---	-65	-65	---	---
	Note 3				Note 2	Note 2		

Dependent: Note 14

Parameter:	Ef	If	ib	Pi	pi	Du	tpc	prr
Units:	V	A	a	W	kw	---	us	pps
© Maximum:	---	---	30.0	720	720	0.001	2.5	4000
Minimum:	---	---		---	---	---	0.25	---
	Note 3	Note 3						

tpc us	rrv kv/us Note 1		tfc us Note 12
	Min.	Max.	Max.
2.5	70	140	0.70
1.5	70	160	0.60
1.0	70	165	0.50
0.25	70	180	0.20

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Mounting position: Any
Support: Mounting flange
Cooling: Forced air required (Note 4)

Magnet: Note 15
Coupling: Waveguide flange UG-148C/U conforming to Specification MIL-F-3922/62
Net weight: 25 pounds maximum

GENERAL:

Qualification - Required

© Denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:17Z
Check the source to verify that this is the current version before use.

Method	Requirement or Test	Conditions	AQL percent defec-tive	Insp. level or code	Symbol	Limits Min.	Limits Max.	units
	Qualification							
1143	Air cooling	Osc. (2); Notes 2, 4 and 5	---	---	AT	---	150	°C.
1031	High frequency vibration	No voltage; t = 60	---	---	- - -	- - -	- - -	- - -
4223	Starting torque		---	---	Torque	---	160	in. -oz
4223	Operating torque		---	---	Torque	---	160	in. -oz
4223	Tuner stop torque		---	---	Torque	---	200	in. -oz
4027	Temperature coefficient	T = 45° to 75°C.: f = 5600 MHz; Note 16 Osc. (1)	---	---	AF AT	---	0.25	MHz/°C
	Quality conformance inspection, part 1							
1301	Heater current	tk = 180 sec. min. Note 3	0.65	II	If	---	---	A
	Oscillation (1)	VSWR = 1.05 max.;						
4303	Heater-cathode warmup time	tk = 180 sec. (max); Note 3	- - -	---	---	---	---	
4304	Pulse characteristics	tp = 0.25 ± 0.03 usec; Du = 0.00024 rrv = 180 kv/usec rein; Note 1	- - -	---	---	---	- - -	
1256	Average anode current	Ib = 5.3 mAdc	- - -	---	- - -	- - -	- - -	
4250	Power output	Note 9	0.65	II	Po	43	---	watts
4308	RF bandwidth	Notes 6 and 7	0.65	II	AF	---	2.0tp	MHz
4308	Minor lobes	Notes 6 and 7	0.65	II	Ratio	8	---	dB
	Oscillation (2)	VSWR = 1.05 max; except as noted						
4303	Heater -cathode warmup time	tk = 180 sec (max); Note 3						

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:17Z
Check the source to verify that this is the current version before use.

Method	Requirement or Test	Conditions	AQL percent defective	Insp. level or code	Symbol	Limits		Units
						Min.	Max.	
	Quality conformance inspection, part 1 cont'd							
4304	Pulse characteristics	tp = 1.30 ± 0.13 usec; Du = 0.00085 rrv = 160 kv/usec rein; Note 1	---	---	---	---	---	---
1256	Average anode current	Ib = 18.7 mAdc;	---	---	---	---	---	---
4306	Pulse voltage	Note 6	0.65	II	epy	20.0	23.0	kv
4307	Power out put	Note 6	0.65	II	Po	150	---	watts
4223	Mechanical tuning range	Upper Limit Lower Limit	0.65 0.65	II II	F F	--- 5450	5825 ---	MHz MHz
4308	RF bandwidth	Notes 6 and 7	0.65	II	F	---	2.0/ tp	MHz
4308	Minor lobes	Notes 6 and 7	0.65	II	Ratio	8	---	dB
4315	Stability	Notes 6, 8 and 16	0.65	II	Missing pulse	---	0.25	%
4310	Pulling factor	Note 6; VSWR = 1.5	0.65	II	AF	---	15	MHz
4311	Pushing factor	Note 6; Ib = 16.5 to 20.5 mAdc	0.65	II	AF	---	2.0	MHz
	Quality conformance inspection, part 2							
1031	Low frequency vibration	No voltage; t = 60 see;	6.5	S3	---	---	---	---
4315	Stability	Osc. (1); Notes 6, 8 and 17	0.65	S3	Missing pulse	---	0.25	%
	Quality conformance inspection, part 3							
---	Life test	Group D; Osc. (2) Frequency = 5450 MHz Note 11; VSWR = 1.5:1 (rein) cycled through λg in 15 min (max) See Note 18			RF time	1200 1000	--- ---	cycles hours

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:17Z
Check the source to verify that this is the current version before use.

Method	Requirement or Test	Conditions	AQL percent defective	Insp. level or code	Symbol	Limits Min.	Limits Max.	Units
	Quality conformance inspection, part 3 cont'd							
---	Life test end points							
	Power output	Osc. (2); Note 6			Po	120	---	Watts
	Tunable frequency	Osc. (2) Upper limit Osc. (2) Lower limit			F F	--- 5450	5825 ---	
	RF bandwidth	Osc. (1) and Osc. (2)			F	---	3.0 tp	
	Minor lobes	Osc. (1) and Osc. (2)			Ratio	6	---	
	Stability	Osc. (2) Notes 6 and 8			Missing pulse	---	0.5	%
	Stability	Osc. (1) Notes 6, 8 and 17			Missing pulse	---	0.5	%

1. The rate of rise of voltage (rrv) shall be expressed in kilovolts per microsecond. Any capacitance used in the viewing system shall not exceed 6 pF.

2. The temperature shall be measured at the point shown on figure 1.

© 3. This magnetron may be manufactured with either of two filament designs. Each tube shall be decaled or stamped with the nominal preheat Ef/If value for its design (i.e., Ef=13.5V/If=2.5A or Ef=11.0V/ If=11.0A). Ratings and limits for the filament are tabulated below:

Requirement or Test	Ef=13.5V/If=2.5A Min	Ef=13.5V/If=2.5A Max	Ef=11.0V/If=11.0A Min	Ef=11.0V/If=11.0A Max	Units
Absolute ratings:					
If (surge)	---	10.0	---	40.0	A
Ef	---	15.5	---	12.5	V
If	---	3.0	---	14.0	A
QCI, Part 1:					
Heater Current					
Ef=13.5V	2.35	2.75	---	---	A
Ef=11.0V	---	---	10.0	12.0	A
Heater/Cathode Warmup:					
Osc (1)					
Preheat Ef	13.5		11.0		V
Operate Ef	13.5		11.0		V
Osc (2)					
Preheat Ef	13.5		11.0		V
Operate Ef	13.0		10.5		V

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:17Z
Check the source to verify that this is the current version before use.

4. With a total airflow of approximately 30 c.f.m. at approximately 760 mm Hg. and 25°C, the temperature shall not exceed the limit specified.

5. The frequency shall be the frequency of minimum power output between 5450 and 5825 MHz.

6. Though the requirement exists continually from 5450 to 5825 MHz tests shall be performed at the following frequencies:

F1 = 5450 ± 5 MHz
F2 = 5625 ± 5 MHz
F3 = 5825 ± 5 MHz

7. The spectrum shall be measured with a VSWR of 1.5 introduced in the load at a distance no greater than 1 meter from the magnetron coupling flange, the phase being adjusted to produce maximum spectrum degradation. A satisfactory spectrum is one whose slope does not change sign more than once for power levels greater than the specified dB below its peak.

8. Stability shall be measured with the tube terminated in a transmission line of 1.5 VSWR adjusted in phase to produce maximum instability. The tube shall be considered stable if the specified missing pulse limit is not exceeded during any 3 minute interval of a 6-minute test period.

9. The power output shall be continuous over the range from 5450 to 5825 MHz. At no frequency in this range shall the power output be less than the specified minimum value.

10. The frequency shall be measured at the extremes of the specified temperature range. The anode temperature shall be measured at the point shown on figure 1.

11. Starting at 5450 MHz the frequency will be increased in 75 MHz increments after each 200 hours (approximately) of total cycle operation.

12. The time of fall of current (tfc) is measured from the zero to 85 percent level of the trailing edge of the current pulse.

13. The ratings have been divided into two groups as follows:

(a) Independent (ratings which may be obtained simultaneously).
(b) Dependent (ratings which are interrelated and may not necessarily be obtained simultaneously).

14. To relate the various parameter, the following formula shall be used:

$$Pi = ib \times Du \times 21500$$

15. In handling and mounting the magnetron, care shall be exercised to prevent demagnetization.

16. This test shall be performed at the conclusion of the holding period.

17. The requirement shall be met with the heater voltage at plus 10 percent and minus 10 percent from the voltage specified for test.

18. Test cycles shall be as follows:

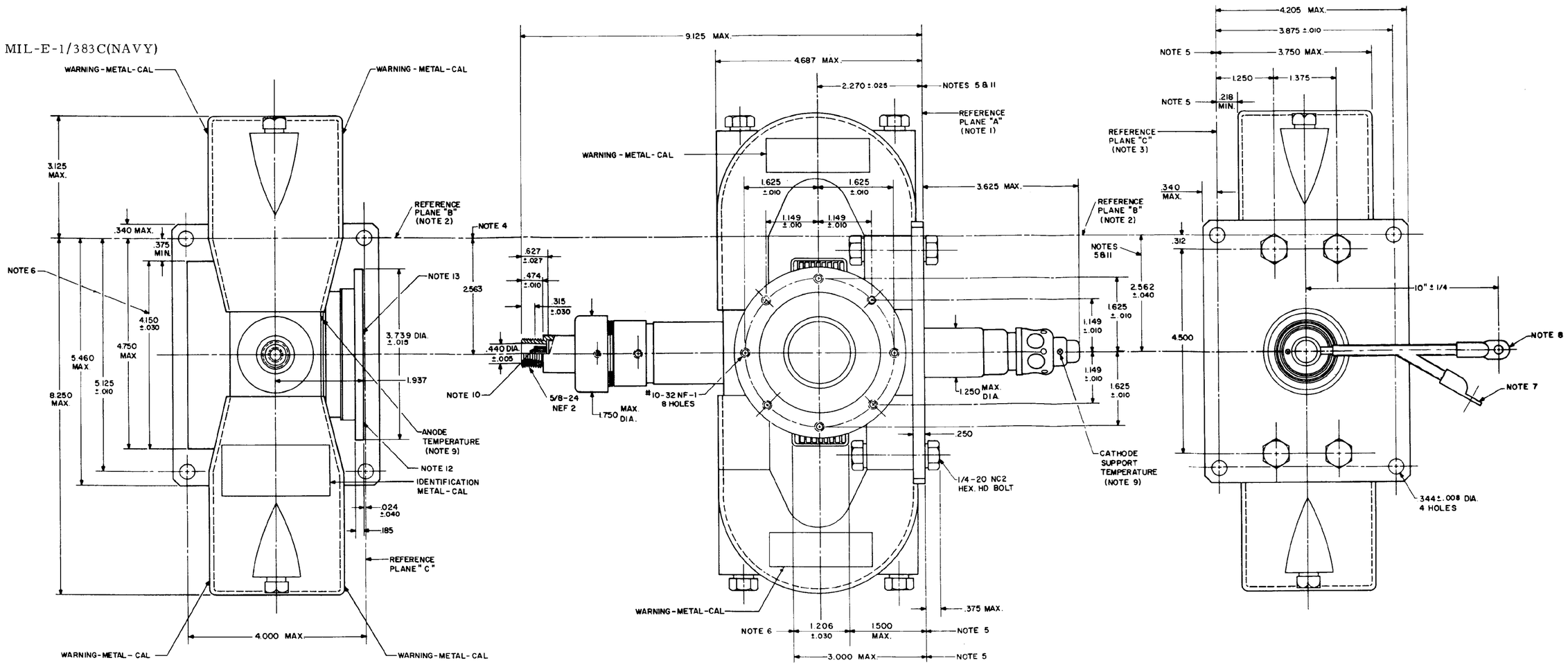
Condition	Ib	Duration
Standby	0 A	3.0 Min. (Max)
Osc. (2)	18.7	50 Min (Min)
Off	0	7.0 Min (Min)

User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

(Project 5960-N010)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:17Z
Check the source to verify that this is the current version before use.



E445

Figure 1 - Outline dimensions of tube type 6344A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:17Z
Check the source to verify that this is the current version before use.

NOTE S:

1. Reference plane "A" lies in top surface of mounting plate.
2. Reference plane "B" perpendicular to "A" passing through the two holes in mounting plate as shown.
3. Reference plane "C" perpendicular to "A" and "B" passing through the two holes in mounting plate as shown.
4. Parts on this axis may deviate from center line by 1/8 inch.
5. Includes angular as well as lateral deviation.
6. This dimension applies to size of radiator only.
7. Common heater-cathode connection.
8. Heater connection.
9. Cathode support and anode temperatures shall be measured at points indicated.
10. Spline requirements:

14-1/2	Pressure angle
48	Pitch
12	Teeth
0.25	Pitch diameter

11. Refers to the centerline of the waveguide as determined by centerline of tapped holes.
12. The output flange shall be parallel to reference plane "C" within 0.025.
13. The glass portion of the output shall not extend beyond the flange.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:17Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:17Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:17Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

☆ U.S. GOVERNMENT PRINTING OFFICE: 1974— 713-155/6578

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:17Z
Check the source to verify that this is the current version before use.